1. 设a，b，c是三角形的三条边的长，A，B，C分别是此三边对应的三个角的量度，求$\frac{\partial A}{\partial a}$，$\frac{\partial A}{\partial b}$，$\frac{\partial A}{\partial c}$.

解：由余弦定理.

$$\cos A=\frac{b^2+c^2-a^2}{2bc}.$$

$$A=\arccos\frac{b^2+c^2-a^2}{2bc}.$$

$$\frac{\partial A}{\partial a}=-\frac{1}{\sqrt{1-\cos^2 A}}\cdot\frac{-2a}{2bc}=\frac{a}{\sin A}\cdot\frac{1}{bc}=\frac{a}{bc\sin A}.$$

$$\frac{\partial A}{\partial b}=-\frac{1}{\sqrt{1-\cos^2 A}}\cdot\frac{2b(2bc)-2c\cdot(b^2+c^2-a^2)}{(2bc)^2}$$

$$=-\frac{1}{\sin A}\cdot\frac{2b-2c\cdot\cos A}{2bc}=\frac{c\cdot\cos A-b}{\sin A}.$$

$$\frac{\partial A}{\partial c}=-\frac{1}{\sqrt{1-\cos^2 A}}\cdot\frac{2c(2bc)-2b\cdot(b^2+c^2-a^2)}{(2bc)^2}.$$

$$=\cancel{b\cdot\sin A}\ \frac{b\cdot\cos A-c}{\sin A}.$$

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2. 设 $z=f\left(y+\varphi(x-y),e^{2x}\right)$，其中 f 有二阶连续偏导数，φ 有二阶导数，求 $\frac{\partial^2 z}{\partial x\partial y}$.

解：$\frac{\partial z}{\partial x}=f_1'\cdot\varphi'+2\cdot e^{2x}\cdot f_2'$

$$\cancel{\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=f_{11}''\cdot\varphi'+f_1'\cdot\varphi''\cdot(-1)+2e^{2x}\cdot f_2''}$$

$$\cancel{\frac{\partial^2 z}{\partial x\partial y}=f_{11}''\cdot\varphi'-f_1'\cdot\varphi''+2\cdot e^{2x}\cdot f_2''}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=f_{11}''\cdot(1-\varphi')\cdot\varphi'+f_1'\cdot\varphi''(-1)+2\cdot e^{2x}f_{21}''\cdot(1-\varphi')$$

$$=\left(f_{11}''\varphi'+2\cdot e^{2x}f_{12}''\right)\cdot(1-\varphi')-f_1'\varphi''$$

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3. 证明函数$y(x,t)=\varphi(x+at)+\varphi(x-at)+\int_{x-at}^{x+at} f(z)\,dz$满足方程$\frac{\partial^2 y}{\partial t^2}=a^2\frac{\partial^2 y}{\partial x^2}$(其中$f$可导，$\varphi$二阶可导).

解：$\frac{\partial y}{\partial x}=\varphi'(x+at)+\varphi'(x-at)+f(x+at)-f(x-at)$

$$\frac{\partial^2 y}{\partial x^2}=\varphi''(x+at)+\varphi''(x-at)+f'(x+at)-f'(x-at)$$

$$\frac{\partial y}{\partial t}=a\cdot\varphi'(x+at)+(-a)\cdot\varphi'(x-at)+a\cdot f'(x+at)-f'(x-at)\cdot(-a)$$

$$=a\cdot\varphi'(x+at)+(-a)\cdot\varphi'(x-at)+a\cdot f'(x+at)+f'(x-at)\cdot a.$$

$$\frac{\partial^2 y}{\partial t^2}=a^2\varphi''(x+at)+a^2\varphi''(x-at)+af''(x+at)-a^2f''(x-at)$$

$$=a^2\left[\varphi''(x+at)+\varphi''(x-at)+f'(x+at)-f'(x-at)\right]$$

$$=a^2\cdot\frac{\partial^2 y}{\partial x^2}.$$

$$\therefore\ \frac{\partial^2 y}{\partial t^2}=a^2\cdot\frac{\partial^2 y}{\partial x^2}.$$

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4. 设函数 $f(u)$ 具有二阶连续导数，而 $z=f\left(e^x\sin y\right)$ 满足方程 $\dfrac{\partial^2 z}{\partial x^2}+\dfrac{\partial^2 z}{\partial y^2}=e^{2x}z$，求 $f(u)$.

解：

$$\frac{\partial z}{\partial x}=f'\sin y\cdot e^x$$

$$\frac{\partial^2 z}{\partial x^2}=f''(e^x\sin y)^2+f'\sin y\cdot e^x$$

$$\frac{\partial z}{\partial y}=f'\cos y\cdot e^x$$

$$\frac{\partial^2 z}{\partial y^2}=f''(e^x\cos y)^2-f'\sin y\cdot e^x$$

$$\therefore \frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=f''\cdot e^{2x}(\sin^2 y+\cos^2 y)=e^{2x}f(u)$$

$$=f''(u)=f(u)$$

$$\therefore \frac{d^2f}{du^2}-f=0$$

特征方程 $r^2=1\Rightarrow r=\pm1$

$$\therefore f=C_1\cdot e^{u}+C_2\cdot e^{-u}.\quad (C_1, C_2\text{为常数})$$

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5. 设$u=u(x)$是由方程组$u=f(x,y)$，$g(x,y,z)=0$，$h(x,z)=0$所确定的函数，其中f，g，h有连续偏导数，且$h_z'\neq 0$，$g_y'\neq 0$，求$\frac{\mathrm{d}u}{\mathrm{d}x}$.

解：$du=f_x'dx+f_y'\cdot dy$.

$$\begin{cases} g_x'+g_y'\cdot\frac{dy}{dx}+g_z'\cdot\frac{dz}{dx}=0 \\ h_x'+h_z'\cdot\frac{dz}{dx}=0 \end{cases}$$

$$\therefore \frac{dz}{dx}=-\frac{h_x'}{h_z'},\quad \frac{dy}{dx}=\frac{g_z'}{g_y'}\cdot\frac{h_x'}{h_z'}-\frac{g_x'}{g_y'}$$

$$\therefore du=\left[f_x'+f_y'\cdot\left(\frac{g_z'}{g_y'}\cdot\frac{h_x'}{h_z'}-\frac{g_x'}{g_y'}\right)\right]\cdot dx$$

$$\therefore \frac{du}{dx}=f_x'+\frac{f_y'\cdot g_z'\cdot h_x'}{g_y'\cdot h_z'}-\frac{f_y'\cdot g_x'}{g_y'}$$

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6. 设函数 $z(x,y)$ 满足 $\begin{cases}\dfrac{\partial z}{\partial x}=-\sin y+\dfrac{1}{1-xy}\\ z(1,y)=\sin y\end{cases}$，求 $z(x,y)$.

解：

$\cancel{\partial z=\int -\sin y+\frac{1}{1-xy}}$

不妨设 $\frac{dz}{dx}=-\sin y+\frac{1}{1-xy}$ · 其中 $\sin y$，y 为常数

$z=(-\sin y)\cdot x+\frac{-1}{y}\cdot\ln(1-xy)+C$.

∴ 代入原题

$z=-x\cdot\sin y-\frac{1}{y}\cdot\ln(1-xy)+C(x,y)$. $C(x,y$ 是

未知表达式

$\cancel{z(1,y)=-\sin y-\frac{1}{y}\cdot\ln(1-xy)+C(x}$

$z(1,y)=-\sin y-\frac{1}{y}\cdot\ln(1-y)+C(x,y)=\sin y$

∴ $C(x,y)=2\sin y+\frac{1}{y}\ln(1-y)$

代入 z.

有 $z=(2-x)\cdot\sin y+\frac{1}{y}\ln\left|\frac{1-y}{1-xy}\right|$.

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7. 设 $f(x,y)$ 有一阶连续偏导数，且 $f(x,x^2)=1$，$f'_x(x,x^2)=x$，求 $f'_y(x,x^2)$.

解：$f(x,x^2)=1$，其中 $y=x^2$

$\therefore f'_x(x,x^2)+f'_y(x,x^2)\cdot\frac{dy}{dx}=0$

$\therefore x+f_y'\cdot(x,x^2)\cdot 2x=0$

$\therefore f_y'=-\frac{1}{2}$

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8. 设 $f(u,v)$ 具有二阶连续偏导数，且满足 $\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}=1$，又 $g(x,y)=f\left(xy,\frac{1}{2}(x^2-y^2)\right)$，求 $\frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial y^2}$.

解：

$g(x,y)=f(u,v)$. $u=xy$, $v=\frac{1}{2}(x^2-y^2)$

$$\frac{\partial g}{\partial x}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}=\frac{\partial f}{\partial u}\cdot y+\frac{\partial f}{\partial v}\cdot x$$

$$\frac{\partial^2 g}{\partial x^2}=\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial x}\cdot y+\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial x}\cdot x+\frac{\partial f}{\partial v}$$

$$=\frac{\partial^2 f}{\partial u^2}\cdot y^2+\frac{\partial^2 f}{\partial v^2}\cdot x^2+\frac{\partial f}{\partial v}$$

$$\frac{\partial g}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=\frac{\partial f}{\partial u}\cdot x+\frac{\partial f}{\partial v}\cdot(-y)$$

$$\frac{\partial^2 g}{\partial y^2}=\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial y}\cdot x+\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial y}\cdot(-y)+(-1)\cdot\frac{\partial f}{\partial v}$$

$$=\frac{\partial^2 f}{\partial u^2}\cdot x^2+\frac{\partial^2 f}{\partial v^2}\cdot y^2-\frac{\partial f}{\partial v}$$

$$\therefore \frac{\partial^2 g}{\partial x^2}+\frac{\partial^2 g}{\partial y^2}=\left(\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}\right)\cdot x^2+\left(\frac{\partial^2 f}{\partial u^2}+\frac{\partial^2 f}{\partial v^2}\right)\cdot y^2$$

$$=x^2+y^2.$$

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9. 作变换$u=x$，$v=x^2-y^2$，求方程$y\frac{\partial z}{\partial x}+x\frac{\partial z}{\partial y}=0$的解.

解: $\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}+\frac{\partial z}{\partial v}\cdot 2x$

$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial v}\cdot -2y$.

$\therefore\ y\cdot\frac{\partial z}{\partial x}+x\cdot\frac{\partial z}{\partial y}=0,\ \frac{\partial z}{\partial u}=0$

$\therefore\ z=f(v)$. f为任意函数,可微函数.

$z=f(x^2-y^2)$.

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10. 设 $z=z(x,y)$ 有二阶连续偏导数，$u=x-ay$，$v=x+ay$，变换方程 $\frac{\partial^2 z}{\partial y^2}=a^2\frac{\partial^2 z}{\partial x^2}$.

由题：$\begin{cases} x=\frac{u+v}{2} \\ y=\frac{v-u}{2a} \end{cases}$

$$\therefore \frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

$$=\frac{1}{2}\cdot\frac{\partial z}{\partial x}+\left(-\frac{1}{2a}\right)\cdot\frac{\partial z}{\partial y}$$

$$\frac{\partial^2 z}{\partial u\partial v}=\frac{1}{4}\cdot\frac{\partial^2 z}{\partial x^2}+\left(-\frac{1}{2a}\right)\cdot\left(\frac{1}{2a}\right)\cdot\frac{\partial^2 z}{\partial y^2}$$

$$=\frac{1}{4}\cdot\left(\frac{\partial^2 z}{\partial x^2}-\frac{1}{a^2}\cdot\frac{\partial^2 z}{\partial y^2}\right)$$

由 $\frac{\partial^2 z}{\partial y^2}=a^2\cdot\frac{\partial^2 z}{\partial x^2}$.

$$\therefore \frac{\partial^2 z}{\partial u\partial v}=0$$

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11. 用变换 $x=e^s$，$y=e^t$ 变换方程 $ax^2\frac{\partial^2 u}{\partial x^2}+2bxy\frac{\partial^2 u}{\partial x\partial y}+cy^2\frac{\partial^2 u}{\partial y^2}=0$（其中 a，b，c 为常数）.

解：$\begin{cases} s=\ln x \\ t=\ln y \end{cases}$

$\therefore \frac{\partial u}{\partial x}=\frac{\partial u}{\partial s}\cdot\frac{\partial s}{\partial x}=\frac{\partial u}{\partial s}\cdot\frac{1}{x}$

$\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial s^2}\cdot\frac{1}{x^2}-\frac{1}{x^2}\cdot\frac{\partial u}{\partial s}$

$\frac{\partial u}{\partial y}=\frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial y}=\frac{\partial u}{\partial s}\cdot\frac{1}{y}$

$\frac{\partial^2 u}{\partial y^2}=\frac{\partial^2 u}{\partial t^2}\cdot\frac{1}{y^2}-\frac{1}{y^2}\cdot\frac{\partial u}{\partial t}$

$\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial\left(\frac{\partial u}{\partial s}\cdot\frac{1}{x}\right)}{\partial t}\cdot\frac{\partial t}{\partial y}$

$=\frac{\partial^2 u}{\partial s\cdot\partial t}\cdot\frac{1}{xy}$

$\therefore$ 代入原式

$a\cdot x^2\cdot\frac{1}{x^2}\left(\frac{\partial^2 u}{\partial s^2}-\frac{\partial u}{\partial s}\right)+2b\cdot xy\cdot\frac{1}{xy}\frac{\partial^2 u}{\partial s\partial t}+c^2\cdot y^2\cdot\frac{1}{y^2}\left(\frac{\partial^2 u}{\partial t^2}-\frac{\partial u}{\partial t}\right)$

$=0.$

~~$\therefore a\cdot\frac{\partial^2 u}{\partial s^2}+2b\cdot\frac{\partial^2 u}{\partial x\partial y}$~~

$\therefore a\cdot\frac{\partial^2 u}{\partial s^2}+2b\cdot\frac{\partial^2 u}{\partial s\partial t}+c^2\cdot\frac{\partial^2 u}{\partial t^2}=a\cdot\frac{\partial u}{\partial s}+c\cdot\frac{\partial u}{\partial t}$

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12. 利用变量代换 $\xi = x+t$，$\eta = x-t$ 求弦振动方程 $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ 的解.

解：

$$\begin{cases} x = \frac{\xi+\eta}{2} \\ t = \frac{\xi-\eta}{2}. \end{cases}$$

$$\frac{\partial u}{\partial \xi} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \xi} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial \xi} = \frac{1}{2}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t}\right)$$

$$\frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial \xi}\right) = \frac{1}{2}\cdot\frac{\partial\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t}\right)}{\partial \eta} = \frac{1}{2}\cdot\left(\frac{\partial^2 u}{\partial x^2}\cdot\frac{\partial x}{\partial \eta} + \frac{\partial^2 u}{\partial t^2}\cdot\frac{\partial t}{\partial \eta}\right)$$

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = \frac{1}{4}\left(\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial t^2}\right)$$

$$\because \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$$

$$\therefore \frac{\partial^2 u}{\partial \xi \partial \eta} = 0$$

可得通解方程为

$u = f(\xi) + g(\eta)$. f, g 为任意可微函数

$u = f(x+t) + g(x-t)$

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13. 设函数 $u=f(x,y,z)$ 在点 $M(x_0,y_0,z_0)$ 可微，又设向量 $\boldsymbol{a}=(1,2,-2)$，$\boldsymbol{b}=(2,1,2)$，$\boldsymbol{c}=(3,4,0)$，已知在点 M 处 $\frac{\partial u}{\partial \boldsymbol{a}}=1$，$\frac{\partial u}{\partial \boldsymbol{b}}=-6$，$\frac{\partial u}{\partial \boldsymbol{c}}=-1$，求 $\mathrm{d}f(x_0,y_0,z_0)$.

解：$|\vec{a}|=3$，$|\vec{b}|=3$，$|\vec{c}|=5$，在 M 点有方程组.

$$\therefore \begin{cases} \frac{\partial u}{\partial x}+2\cdot\frac{\partial u}{\partial y}-2\frac{\partial u}{\partial z}=3 \\ 2\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+2\frac{\partial u}{\partial z}=-18 \\ 3\frac{\partial u}{\partial x}+4\cdot\frac{\partial u}{\partial y}=-5 \end{cases}$$

可得矩阵

$$\begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 1 & 2 & -18 \\ 3 & 4 & 0 & -5 \end{bmatrix} \xrightarrow[\text{变换}]{\text{初等行}} \begin{bmatrix} 1 & 2 & -2 & 3 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -15 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\therefore \begin{cases} \frac{\partial u}{\partial x}=-15 \\ \frac{\partial u}{\partial y}=10 \\ \frac{\partial u}{\partial z}=1 \end{cases}$$

$$\therefore df=\frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}\cdot dy+\frac{\partial u}{\partial z}dz$$

$$=-15dx+10dy+dz.$$

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14. 设$u=xyze^{x+y+z}$，求$\dfrac{\partial^9 u}{\partial x^2\partial y^3\partial z^4}$.

解：$\dfrac{\partial u}{\partial z}=xy\cdot(z+1)\cdot e^{x+y+z}$

$\dfrac{\partial^2 u}{\partial z^2}=xy\cdot(z+2)\cdot e^{x+y+z}$

$\dfrac{\partial^3 u}{\partial z^3}=xy\cdot(z+3)\cdot e^{x+y+z}$

$\dfrac{\partial^4 u}{\partial z^4}=xy(z+4)\cdot e^{x+y+z}$

$\dfrac{\partial^5 u}{\partial y\partial z^4}=x\cdot(y+1)\cdot(z+4)\cdot e^{x+y+z}$

$\dfrac{\partial^6 u}{\partial y^2\partial z^4}=x(y+2)\cdot(z+4)\cdot e^{x+y+z}$

$\dfrac{\partial^7 u}{\partial y^3\cdot\partial z^4}=x(y+3)\cdot(z+4)\cdot e^{x+y+z}$

$\dfrac{\partial^8 u}{\partial x\cdot\partial y^3\cdot\partial z^4}=(x+1)(y+3)\cdot(z+4)\cdot e^{x+y+z}$

$\dfrac{\partial^9 u}{\partial x^2\cdot\partial y^3\partial z^4}=(x+2)(y+3)(z+4)\cdot e^{x+y+z}$

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15. 设 $f(x)$，$g(x)$ 是可微函数，且满足 $\begin{cases} u(x,y)=f(2x+5y)+g(2x-5y) \\ u(x,0)=\sin 2x \\ u'_y(x,0)=0 \end{cases}$. 求 $f(x)$，$g(x)$ 及 $u(x,y)$ 的表达式.

解：$\frac{\partial u}{\partial y} = f_y' \cdot \frac{\partial f}{\partial y} + g_y' \cdot \frac{\partial g}{\partial y} = 0$

$= 5(f_y' - g_y') = 0$

$\therefore f_y' = g_y'$

$\because u(x,0) = f(2x) + g(2x) = \sin 2x$

即 $f(u) + g(u) = \sin u$.

$\because f_y' = g_y'$

$\therefore f(x,y) + C_1 = g(x,y) + C_2$.

$f(x,y) + g(x,y) = \sin(x,y)$

即 $f(u) + C_1 = g(u) + C_2$.

$f(u) + g(u) = \sin u$

可得. $\frac{1}{2}\sin u = g(u) + \frac{C_2 - C_1}{2}$

$\frac{1}{2}\sin u = f(u) + \frac{C_1 - C_2}{2}$.

令 $C = \frac{C_2 - C_1}{2}$

$\therefore f(x) = \frac{1}{2}\sin x + C$.

$g(x) = \frac{1}{2}\sin x - C$.

$\therefore u(x,y) = \frac{1}{2}\sin(2x+5y) + \frac{1}{2}\sin(2x-5y)$

$= \sin 2x \cos 5y$ （和差化积）

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16.

(1) 若可微函数 $z=f(x,y)$ 满足方程 $x\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=0$，证明 $f(x,y)$ 在极坐标下只是 θ 的函数.

(2) 若可微函数 $z=f(x,y)$ 满足方程 $\dfrac{\frac{\partial z}{\partial x}}{x}=\dfrac{\frac{\partial z}{\partial y}}{y}$，证明 $f(x,y)$ 只是 ρ 的函数，其中 $\rho=\sqrt{x^2+y^2}$.

解：(1)

$$\begin{cases}\theta=\arctan\frac{y}{x}\\ \rho=\sqrt{x^2+y^2}\end{cases}$$

$$\therefore \frac{\partial z}{\partial x}=\frac{\partial z}{\partial \theta}\cdot\frac{\partial \theta}{\partial x}+\frac{\partial z}{\partial \rho}\cdot\frac{\partial \rho}{\partial x}$$

$$=-\frac{y}{x^2+y^2}\cdot\frac{\partial z}{\partial \theta}+\frac{x}{\sqrt{x^2+y^2}}\cdot\frac{\partial z}{\partial \rho}$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial \theta}\cdot\frac{\partial \theta}{\partial y}+\frac{\partial z}{\partial \rho}\cdot\frac{\partial \rho}{\partial y}$$

$$=\frac{x}{x^2+y^2}\cdot\frac{\partial z}{\partial \theta}+\frac{y}{\sqrt{x^2+y^2}}\cdot\frac{\partial z}{\partial \rho}$$

$$\therefore x\cdot\frac{\partial z}{\partial x}+y\cdot\frac{\partial z}{\partial y}=\frac{x^2+y^2}{\sqrt{x^2+y^2}}\cdot\frac{\partial z}{\partial \rho}=\rho\cdot\frac{\partial z}{\partial \rho}=0$$

$\therefore \dfrac{\partial z}{\partial \rho}=0$. $\therefore z=g(\theta)$. 故 $f(x,y)=g(\theta)$，只为 θ 函数.（g 为任意可微函数）

(2) $\dfrac{\partial z}{\partial \theta}=\dfrac{\partial z}{\partial x}\cdot\dfrac{\partial x}{\partial \theta}+\dfrac{\partial z}{\partial y}\cdot\dfrac{\partial y}{\partial \theta}$ 其中 $\begin{cases}x=\rho\cdot\cos\theta\\ y=\rho\cdot\sin\theta\end{cases}$

$$\therefore \frac{\partial z}{\partial \theta}=\rho\cdot\frac{\partial z}{\partial x}\cdot(-\sin\theta)+\rho\cdot\cos\theta\cdot\frac{\partial z}{\partial y}$$

$$=-y\cdot\frac{\partial z}{\partial x}+x\cdot\frac{\partial z}{\partial y}.$$

$\because \dfrac{\frac{\partial z}{\partial x}}{x}=\dfrac{\frac{\partial z}{\partial y}}{y}$ $\therefore x\dfrac{\partial z}{\partial y}-y\cdot\dfrac{\partial z}{\partial x}=0$

$\therefore \dfrac{\partial z}{\partial \theta}=0$. $\therefore z=h(\rho)$. h 是可微函数.

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17. 设 $u=u(x,y,z)$ 是可微函数，若 $xu'_x+yu'_y+zu'_z=0$，证明在球坐标下 u 仅为 θ，φ 的函数.

解：

$$\begin{cases} z=r\cdot\cos\theta \\ y=r\cdot\sin\theta\cdot\sin\varphi \\ x=r\cdot\sin\theta\cdot\cos\varphi \end{cases} \Rightarrow \begin{cases} r=\sqrt{x^2+y^2+z^2} \\ \theta=\arccos\frac{z}{r} \\ \varphi=\arctan\frac{y}{x} \end{cases}$$

$$\therefore \frac{\partial u}{\partial x}=\frac{x}{\sqrt{x^2+y^2+z^2}}\cdot\frac{\partial u}{\partial r}+\frac{1}{\sqrt{1-\frac{z^2}{x^2+y^2+z^2}}}\cdot\frac{xz}{(x^2+y^2+z^2)^{\frac{3}{2}}}\cdot\frac{\partial u}{\partial\theta}-\frac{y}{x^2+y^2}\cdot\frac{\partial u}{\partial\varphi}$$

$$=\frac{x}{\sqrt{x^2+y^2+z^2}}\frac{\partial u}{\partial r}+\frac{1}{\sin\theta}\cdot\frac{xz}{(x^2+y^2+z^2)^{\frac{3}{2}}}\cdot\frac{\partial u}{\partial\theta}-\frac{y}{x^2+y^2}\cdot\frac{\partial u}{\partial\varphi}$$

$$\frac{\partial u}{\partial y}=\frac{y}{\sqrt{x^2+y^2+z^2}}\cdot\frac{\partial u}{\partial r}+\frac{1}{\sin\theta}\cdot\frac{y\cdot z}{(x^2+y^2+z^2)^{\frac{3}{2}}}\frac{\partial u}{\partial\theta}+\frac{x}{x^2+y^2}\cdot\frac{\partial u}{\partial\varphi}$$

$$\frac{\partial u}{\partial z}=\frac{z}{\sqrt{x^2+y^2+z^2}}\cdot\frac{\partial u}{\partial r}-\frac{1}{\sin\theta}\cdot\frac{x^2+y^2}{(x^2+y^2+z^2)^{\frac{3}{2}}}\cdot\frac{\partial u}{\partial\theta}$$

$$\therefore x\cdot\frac{\partial u}{\partial x}+y\cdot\frac{\partial u}{\partial y}+z\cdot\frac{\partial u}{\partial z}=0$$

$$=\frac{x^2+y^2+z^2}{\sqrt{x^2+y^2+z^2}}\cdot\frac{\partial u}{\partial r}+\frac{1}{\sin\theta}\cdot\left(\frac{x^2z+y^2z-(x^2+y^2)z}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right)\cdot\frac{\partial u}{\partial\theta}$$

$$+\left(-\frac{xy}{x^2+y^2}+\frac{xy}{x^2+y^2}\right)\cdot\frac{\partial u}{\partial\varphi}$$

$$=r\cdot\frac{\partial u}{\partial r}=0.$$

$\therefore \frac{\partial u}{\partial r}=0$，$u=f(\theta,\varphi)$ f 为可微函数

$\therefore$ u 为 θ,φ 的函数.

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18. 如果对所有t，x，y，有$f(tx,ty)=t^n f(x,y)$，则$f(x,y)$叫做n次齐次函数.

设$f(x,y)$是n次齐次函数，且有二阶连续偏导数，证明：

(1) $x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}=nf(x,y)$(此结论称为欧拉定理);

(2) $x^2\frac{\partial^2 f}{\partial x^2}+2xy\frac{\partial^2 f}{\partial x\partial y}+y^2\frac{\partial^2 f}{\partial y^2}=n(n-1)f(x,y)$.

解：$\frac{d f(tx,ty)}{dt}=\frac{\partial f}{\partial(tx)}\cdot\frac{\partial(tx)}{\partial t}+\frac{\partial f}{\partial(ty)}\cdot\frac{\partial(ty)}{\partial y}$

$=x\cdot\frac{\partial f}{\partial(tx)}+y\cdot\frac{\partial f}{\partial(ty)}$

$\frac{d\, t^n\cdot f(x,y)}{dt}=n\cdot t^{n-1}f(x,y)$

$\because f(tx,ty)=t^n\cdot f(x,y)$

故 $\frac{d f(tx,ty)}{dt}=\frac{d\,t^n f(x,y)}{dt}$

$\therefore$ 令$t=1$，有 $x\cdot\frac{\partial f}{\partial x}+y\cdot\frac{\partial f}{\partial y}=n\cdot f(x,y)$

(2). $\frac{d^2f}{dt^2}=\left(\frac{\partial^2 f}{\partial(tx)^2}\cdot\frac{\partial(tx)}{\partial t}+\frac{\partial^2 f}{\partial(tx)\partial(ty)}\cdot\frac{\partial(ty)}{\partial t}\right)\cdot x$

$+\left(\frac{\partial^2 f}{\partial(ty)^2}\cdot\frac{\partial(ty)}{\partial t}+\frac{\partial^2 f}{\partial(tx)\partial(ty)}\cdot\frac{\partial(tx)}{\partial t}\right)\cdot y$

$=x^2\cdot\frac{\partial^2 f}{\partial(tx)^2}+2xy\cdot\frac{\partial^2 f}{\partial(tx)\partial(ty)}+y^2\cdot\frac{\partial^2 f}{\partial(ty)^2}$

$\frac{d^2(t^n f)}{dt^2}=n(n-1)\cdot t^{n-2}\cdot f(x,y)$

令$t=1$，有 $\frac{d^2f}{dt^2}=\frac{d^2(t^n f)}{t^2}$

即 $x^2\cdot\frac{\partial^2 f}{\partial x^2}+2xy\cdot\frac{\partial^2 f}{\partial x\partial y}+y^2\cdot\frac{\partial^2 f}{\partial y^2}=n(n-1)\cdot f(x,y)$

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19. 设 $f(x,y)=\begin{cases}\dfrac{x^2y^2}{\left(x^2+y^2\right)^{\frac{3}{2}}} & x^2+y^2\neq 0\\ 0 & x^2+y^2=0\end{cases}$，证明在点 $(0,0)$ 处 $f(x,y)$ 连续且偏导数存在，但不可微.

解：$f_x'(0,0)=\lim\limits_{\Delta x\to 0}\dfrac{f(x+\Delta x,0)-f(0,0)}{\Delta x}=0$

$f_y'(0,0)=\lim\limits_{\Delta y\to 0}\dfrac{f(0,y+\Delta y)-f(0,0)}{\Delta y}=0$

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\Delta z=\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\Delta x^2\Delta y^2}{(\Delta x^2+\Delta y^2)^{\frac{3}{2}}}\leq\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\Delta x^2\Delta y^2}{2\Delta x^{\frac{3}{2}}\cdot\Delta y^{\frac{3}{2}}}$$

$$=\frac{1}{2}\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\sqrt{\Delta x\Delta y}=0$$

$\because \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\Delta z\geq 0 \quad \therefore \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\Delta z=0$

又 $\because \lim\limits_{\substack{\Delta z\to 0\\ \rho\to 0}}\dfrac{\Delta z}{\rho}=\dfrac{\Delta x^2\Delta y^2}{(\Delta x^2+\Delta y^2)^2}$

令 $\Delta y=k\cdot\Delta x$

$\lim\limits_{\substack{\Delta z\to 0\\ \rho\to 0}}\dfrac{\Delta z}{\rho}=\dfrac{k^2\Delta x^4}{\Delta x^4\cdot(1+k^2)^2}=\dfrac{k^2}{(1+k^2)^2}\neq 0$

故 $f(x,y)$ 在 $(0,0)$ 点连续且偏导存在

但不可微.

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20. 设函数$u=f\left(\ln\sqrt{x^2+y^2}\right)$满足方程$\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}=\left(x^2+y^2\right)^{\frac{3}{2}}$，且$\lim\limits_{x\to 0}\dfrac{\int_0^1 f(xt)\,dt}{x}=-1$，求$f$的表达式.

解: $\dfrac{\partial u}{\partial x}=f'\cdot\dfrac{x}{x^2+y^2}$ $\quad\dfrac{\partial u}{\partial y}=f'\dfrac{y}{x^2+y^2}$

$\cancel{\dfrac{\partial^2 u}{\partial x^2}=f''\dfrac{x^2}{(x^2+y^2)^2}}$

$$\frac{\partial^2 u}{\partial x^2}=f''\frac{x^2}{(x^2+y^2)^2}+f'\frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2}=f''\cdot\frac{y^2}{(x^2+y^2)^2}+f'\cdot\frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial x^2}=f''\cdot\frac{x^2+y^2}{(x^2+y^2)^2}=(x^2+y^2)^{\frac{3}{2}}$$

$\therefore f''(u)=(x^2+y^2)^{\frac{5}{2}}$

其中 $u=\ln\sqrt{x^2+y^2}$，$x^2+y^2=e^{2u}$

$\therefore f''(u)=e^{5u}$.

$\therefore f'(u)=\frac{1}{5}\cdot e^{5u}+C_1$

$f(u)=\frac{1}{25}e^{5u}+C_1\cdot u+C_2$

$\therefore f(x)=\frac{1}{25}e^{5x}+C_1\cdot x+C_2$.

$F(x)=\frac{1}{125}\cdot e^{5x}+\frac{C_1}{2}x^2+C_2x+C_3$

$\int_0^1 f(xt)\,dt=\frac{1}{x}\int_0^x f(u)\,du$

$$\therefore \lim_{x\to 0}\frac{\int_0^1 f(xt)\,dt}{x}=\lim_{x\to 0}\frac{\frac{1}{125}e^{5x}+\frac{C_1}{2}x^2+C_2x-\frac{1}{125}}{x^2}=-1$$

$\cancel{=\lim_{x\to 0}\left(\frac{C_1}{2}+\frac{\frac{1}{125}e^{5x}-\frac{1}{125}+C_2x}{x^2}\right)}$

$$\therefore \lim_{x\to 0}\frac{\frac{1}{25}e^{5x}+C_1x+C_2}{2x}=-1$$

$$\lim_{x\to 0}\frac{\frac{1}{5}e^{5x}+C_1}{2}=-1$$

$\therefore C_1=-\frac{11}{5}$，$C_2=-\frac{1}{25}$

$$\therefore f(x)=\frac{1}{25}e^{5x}-\frac{11}{5}x-\frac{1}{25}$$

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第十节 综合例题

21. 证明曲线 $x=e^t\cos t$，$y=e^t\sin t$，$z=e^t$ 与圆锥面 $x^2+y^2=z^2$ 的所有母线以等角相交.

解：曲线的切向量 $\vec{s}=\left\{\frac{dx}{dt},\frac{dy}{dt},\frac{dz}{dt}\right\}$

$$=\left\{e^t(\cos t-\sin t),\ e^t(\sin t+\cos t),\ e^t\right\}$$

$$=z\cdot\left\{(\cos t-\sin t),(\sin t+\cos t),1\right\}$$

圆锥面如图，母线方向为 $\vec{l}=\{z\cdot\cos t, z\cdot\sin t, z\}=z\{\cos t,\sin t,1\}$

（两曲线相交，故参数 t 一致）

∴设 $\vec{e_1}=\{(\cos t-\sin t),(\sin t+\cos t),1\}$

$\vec{e_2}=\{\cos t,\sin t,1\}$

只需证明，$\vec{e_1}$ 与 $\vec{e_2}$ 夹角不变，即证原命题

设夹角为 θ

$\cancel{\cos\theta}\ \cos\theta=\frac{\vec{e_1}\cdot\vec{e_2}}{|\vec{e_1}|\cdot|\vec{e_2}|}=\frac{2}{\sqrt{3}\cdot\sqrt{2}}=\frac{2}{\sqrt{6}}=\frac{\sqrt{6}}{3}$

证毕

22. 证明 $z=xf\left(\frac{y}{x}\right)$ 曲面上任何一点的切平面通过一定点.

解：设 $P(x_0,y_0,z_0)$

$\therefore F=x\cdot f\left(\frac{y}{x}\right)-z$

~~$F_x'=f(\frac{y_0}{x_0})$~~

$F_x'=f\left(\frac{y}{x}\right)+x\cdot f'\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right)$

$\quad=f\left(\frac{y}{x}\right)-\frac{y}{x}\cdot f'\left(\frac{y}{x}\right)$

$F_y'=f'\left(\frac{y}{x}\right)$

$F_z'=-1$

$\therefore$ 在P点法向量

$\vec{n}=\left\{f\left(\frac{y_0}{x_0}\right)-\frac{y_0}{x_0}\cdot f'\left(\frac{y_0}{x_0}\right),\ f'\left(\frac{y_0}{x_0}\right),\ -1\right\}$

$\therefore$ 过P点，以 $\vec{n}$ 为法向量平面方程.

$\left[f\left(\frac{y_0}{x_0}\right)-\frac{y_0}{x_0}f'\left(\frac{y_0}{x_0}\right)\right]\cdot(x-x_0)+f'\left(\frac{y_0}{x_0}\right)\cdot(y-y_0)-(z-z_0)=0$

$\because x_0f\left(\frac{y_0}{x_0}\right)=z_0$

$\therefore$ 有 $\left[f\left(\frac{y_0}{x_0}\right)-\frac{y_0}{x_0}f'\left(\frac{y_0}{x_0}\right)\right]\cdot x+f'\left(\frac{y_0}{x_0}\right)\cdot y-z=0$

故 $(0,0,0)$ 满足上述方程

证毕

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第十节 综合例题

23. 求曲面 $x=u+v$，$y=u^2+v^2$，$z=u^3+v^3$ 在 $u=1$，$v=-1$ 处的切平面方程.

解：$\begin{cases} x=u+v \\ y=u^2+v^2 \\ z=u^3+v^3 \end{cases}$

对x求偏导

$\begin{cases} 1=\frac{\partial u}{\partial x}+\frac{\partial v}{\partial x} \\ 0=2u\cdot\frac{\partial u}{\partial x}+2v\cdot\frac{\partial v}{\partial x} \\ \frac{\partial z}{\partial x}=3u^2\cdot\frac{\partial u}{\partial x}+3v^2\frac{\partial v}{\partial x}. \end{cases}$

对y求偏导

$\begin{cases} 0=\frac{\partial u}{\partial y}+\frac{\partial v}{\partial y} \\ 1=2u\cdot\frac{\partial u}{\partial y}+2v\cdot\frac{\partial v}{\partial y} \\ \frac{\partial z}{\partial y}=3u^2\cdot\frac{\partial u}{\partial y}+3v^2\frac{\partial v}{\partial y} \end{cases}$

将 $u=1$, $v=-1$ 代入上式

可得 $\frac{\partial z}{\partial x}=3$，$\frac{\partial z}{\partial y}=0$

$\therefore \vec{n}=\{z_x', z_y', -1\}=\{3,0,-1\}$

$\because$ 当 $u=1$, $v=-1$ 时，

$\begin{cases} x=0 \\ y=2 \\ z=0 \end{cases}$

$\therefore 3(x-0)+0+(-1)\cdot z=0$

$\therefore 3x=z$.

$\therefore$ 切平面方程 $3x=z$.

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第十节　综合例题

24. 试求一平面，使它通过曲线 $\begin{cases} y^2 = x \\ z = 3(y-1) \end{cases}$ 在 $y=1$ 处的切线，且与曲面 $x^2+y^2=4z$ 相切.

解：~~$\because x^2+y^2=4z$~~ 当 $y=1$ 时，$x=1, z=0$

$z=\frac{1}{4}(x^2+y^2)$

$\frac{\partial z}{\partial x}=\frac{1}{2}x$，$\frac{\partial z}{\partial y}=\frac{1}{2}y$

~~当 $y=1$ 时，其曲面的法向量 $\vec{n}=\{\frac{1}{2},\frac{1}{2},-1\}$~~

曲面的法向量为 $\vec{n}=\{\frac{1}{2}x,\frac{1}{2}y,-1\}$

$\begin{cases} y^2=x \\ z=3(y-1) \end{cases}$

$\therefore \frac{dx}{dy}=2y$，$\frac{dz}{dy}=3$

在 $y=1$ 处，$\left.\frac{dx}{dy}\right|_{y=1}=2$，$\frac{dz}{dy}=3$

$\therefore$ 在点 $(1,1,0)$ 处，切向量 $\vec{s}=(2,1,3)$

$\therefore$ 切点处法向量 $\vec{n}=\{\frac{1}{2}x_0,\frac{1}{2}y_0,-1\}$

$\therefore \begin{cases} \frac{1}{2}x_0(x_0-1)+\frac{1}{2}y_0(y_0-1)-z=0 \\ \vec{n}\cdot\vec{s}=x+\frac{1}{2}y-3=0 \\ x^2+y^2=4z \end{cases}$

可解得 $\begin{cases} x=2 \\ y=2 \end{cases}$ 或 $\begin{cases} x=\frac{12}{5} \\ y=\frac{6}{5} \end{cases}$

~~$\therefore \vec{n}=\{1,2,$~~

$\therefore \vec{n}=\{1,1,-1\}$ 或 $\vec{n}=\{\frac{6}{5},\frac{3}{5},-1\}$

代入点 $(1,1,0)$ 由点法式得

$x+y-z=2$，或 $6x+3y-5z=9$

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第十节　综合例题

25. 求曲面 $z=xy$ 的法线，使它与平面 $x+3y+z+9=0$ 垂直.

解: $\frac{\partial z}{\partial x}=y$, $\frac{\partial z}{\partial y}=x$

设平面 $x+3y+z+C=0$ 与曲线切于

$P(x_0, y_0, z_0)$

∴ 该处法向量 $\vec{n}=\{\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, -1\}$

∴ $\frac{y_0}{1}=\frac{x_0}{3}=\frac{-1}{1}$

∴ $\begin{cases} x_0=-3 \\ y_0=-1 \\ z_0=3 \end{cases}$

∴ 该法线直线方程

$\frac{x+3}{1}=\frac{y+1}{3}=z-3$

∴ $x+3=\frac{y+1}{3}=z-3$

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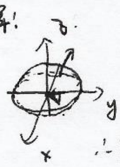
26. 设 $\boldsymbol{n}$ 是曲面 $2x^2+3y^2+z^2=6$ 在点 $P(1,1,1)$ 处指向外侧的法向量，求函数 $u=\dfrac{\sqrt{6x^2+8y^2}}{z}$ 在点 P 处沿方向 $\boldsymbol{n}$ 的方向导数.

$\partial F=2x^2+3y^2+z^2-6$

解：



$F_x'=4x$, $F_x'|_{x=1}=4$

$F_y'=6y$; $F_y'|_{y=1}=6$

$F_z'=2z$; $F_z'|_{z=1}=2$

$\therefore P(1,1,1)$ 的法向量 $\vec{n}=\{2,3,1\}$

$\therefore \vec{e}=\dfrac{\vec{n}}{|\vec{n}|}=\left\{\dfrac{2}{\sqrt{14}},\dfrac{3}{\sqrt{14}},\dfrac{1}{\sqrt{14}}\right\}$

$$\frac{\partial u}{\partial x}=\frac{6x}{\sqrt{6x^2+8y^2}}$$

$$\frac{\partial u}{\partial y}=\frac{8y}{\sqrt{6x^2+8y^2}}$$

$$\frac{\partial u}{\partial z}=-\frac{\sqrt{6x^2+8y^2}}{z^2}$$

在 P 点 $(1,1,1)$ 有 $\dfrac{\partial u}{\partial x}=\left\{\dfrac{6}{\sqrt{14}},\dfrac{8}{\sqrt{14}},-\sqrt{14}\right\}$

$$\therefore \left\{\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right\}\cdot\begin{pmatrix}2\\3\\1\end{pmatrix}\cdot\frac{1}{\sqrt{14}}$$

$$=\frac{12+24}{14}-1=\frac{22}{14}=\frac{11}{7}$$

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27. 求正数λ的值，使得曲面$xyz=\lambda$与曲面$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$在某一点相切.

解：$F=xyz-\lambda$.　　$G=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1=0$

$F_x'=yz$　　$G_x'=\frac{2x}{a^2}$

$F_y'=xz$　　$G_y'=\frac{2y}{b^2}$

$F_z'=xy$.　　$G_z'=\frac{2z}{c^2}$.

$$\begin{cases} G_x'=\mu\cdot F_x' \\ G_y'=\mu\cdot F_y' \\ G_z'=\mu\cdot F_z' \\ F=0 \\ G=0. \end{cases} \Rightarrow \begin{cases} \frac{2x^2}{a^2}=\lambda\cdot\mu \\ \frac{2y^2}{b^2}=\lambda\cdot\mu \\ \frac{2z^2}{b^2}=\lambda\cdot\mu. \\ G=0 \\ F=0 \end{cases}$$

$\therefore \frac{3}{2}\lambda\mu=1,\ \mu=\frac{2}{3\lambda}$

$$\therefore \begin{cases} \frac{x^2}{a^2}=\frac{1}{3} \\ \frac{y^2}{b^2}=\frac{1}{3} \\ \frac{z^2}{c^2}=\frac{1}{3} \end{cases} \Rightarrow \begin{cases} x=\sqrt{\frac{1}{3}}\cdot a \\ y=\sqrt{\frac{1}{3}}\cdot b \\ z=\sqrt{\frac{1}{3}}\cdot c. \end{cases}$$

$\therefore \lambda=xy\cdot z=\frac{\sqrt{3}}{9}abc$.

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$\{0,0,1\}$

28. 求常数a，b，c的值，使函数$f(x,y,z)=axy^2+byz+cx^3z^2$在点$(1,2,-1)$处沿$z$轴正方向的方向导数有最大值64.

解：$\frac{\partial f}{\partial x}=ay^2+cz^2\cdot 3x^2$

$\frac{\partial f}{\partial y}=2axy+bz$

$\frac{\partial f}{\partial z}=2cx^3z+by$

在$(1,2,-1)$处有

$$\begin{cases}\frac{\partial f}{\partial x}=4a+3c\\ \frac{\partial f}{\partial y}=4a-b\\ \frac{\partial f}{\partial z}=2b-2c\end{cases}$$

有最大值意味$\{\frac{\partial f}{\partial x},\frac{\partial f}{\partial y},\frac{\partial f}{\partial z}\}$与$\{0,0,1\}$同向

$$\therefore\begin{cases}4a+3c=0\\ 4a-b=0\\ 2b-2c=64\end{cases}\Rightarrow\begin{cases}a=6\\ b=24\\ c=-8\end{cases}$$

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第十节　综合例题

29. 设有一平面温度场 $T(x,y)=100-x^2-2y^2$，场内有一粒子从 $A(4,2)$ 处出发始终沿着温度上升最快的方向运动，试建立粒子运动所应满足的微分方程，并求出粒子运动的路径方程.

解：由题

$\cancel{dT=\frac{\partial T}{\partial x}dx}$

$\frac{\partial T}{\partial x}=-2x$

$\frac{\partial T}{\partial y}=-4y$

$\therefore \mathrm{grad}\,T=\left(\frac{\partial T}{\partial x},\frac{\partial T}{\partial y}\right)=\{-2x,-4y\}$

∴ 沿温度上升最快的方向 $\{dx,dy\}\parallel\{-2x,-4y\}$

$\therefore \frac{dx}{2x}=\frac{dy}{4y}$

$\therefore 2y\,dx-x\,dy=0$

$\therefore \ln|x^2|=\ln|y|+c$

$\therefore y=c\cdot x^2$

代入 $(4,2)$ 有

$y=\frac{1}{8}x^2,\ (0\le x\le 4)$

30. 已知平面上两定点 $A(1,3)$，$B(4,2)$，试在曲线 $\frac{x^2}{9}+\frac{y^2}{4}=1\ (x\ge 0, y\ge 0)$ 上求一点 C，使 $\triangle ABC$ 的面积最大.

解：$\vec{AB}=\{3,-1\}$，$|\vec{AB}|=\sqrt{9+1}=\sqrt{10}$，AB方程 $x+3y-10=0$

设 $P=(x_0,y_0)$ 在 $\frac{x^2}{9}+\frac{y^2}{4}=1$ 上.

$$d=\frac{|x_0+3y_0-10|}{\sqrt{10}}$$

由图知．曲线向AB侧凹

故最值点在边界处

$$\therefore d_1=\left|\frac{0+6-10}{\sqrt{10}}\right|=\frac{4}{\sqrt{10}}$$

$$d_2=\left|\frac{3+0-10}{\sqrt{10}}\right|=\frac{7}{\sqrt{10}}$$

$\therefore d_2>d_1$

$\therefore$ C为 $(3,0)$ 时，$S_{\triangle ABC}$ 面积最大.

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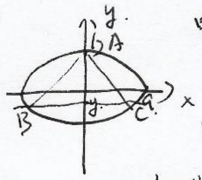
第十节 综合例题

31. 在椭圆$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$内作底边平行于$x$轴的内接三角形，求此类三角形面积的最大值.

解：



由题意如图

必有$y<0$

$A(0,b)$

$BC=2a\cdot\sqrt{1-\frac{y^2}{b^2}}=2\frac{a}{b}\sqrt{b^2-y^2}$

$h=b-y$

$\therefore S_{\triangle ABC}=\frac{1}{2}h\cdot BC$

$=(b-y)\cdot\frac{a}{b}\cdot\sqrt{b^2-y^2}$

$=\frac{a}{b}\cdot\sqrt{(b-y)^3\cdot(b+y)}$

设 $f(y)=(b-y)^3(b+y)$

$f'(y)=(b-y)^2(4b+2y)$

经判定 $f'(y)$在 $y=-\frac{1}{2}b$处

有极大值 $\frac{27}{16}\cdot b^4$

$\therefore S_{\triangle ABC}=\frac{3\sqrt{3}}{4}\cdot ab$

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第十节 综合例题

32. 设 $P(x_1,y_1)$ 是椭圆 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 外的一点，若 $Q(x_2,y_2)$ 是椭圆上离 P 最近的一点，证明 PQ 是椭圆的法线.

解：$d^2=(x-x_1)^2+(y-y_1)^2$.

构造拉格朗日函数.

则 $F=(x-x_1)^2+(y-y_1)^2+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)$

$$\therefore \begin{cases} F_x'=2(x-x_1)+\dfrac{2\lambda x}{a^2}=0 \\ F_y'=2(y-y_1)+\dfrac{2\lambda y}{b^2}=0 \\ \dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1 \end{cases}$$

令 $f=\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$\frac{\partial f}{\partial x}=\frac{2x}{a^2}$，$\frac{\partial f}{\partial y}=\frac{2y}{b^2}$.

$\therefore$ 法线向量 $\vec{n}=\left\{\frac{2x}{a^2},\frac{2y}{b^2}\right\}$

$$\overrightarrow{QP}=\{x-x_1,y-y_1\}=\left\{-\frac{\lambda}{a^2}x,-\frac{\lambda}{b^2}y\right\}$$

$$=-\frac{\lambda}{2}\left\{\frac{2x}{a^2},\frac{2y}{b^2}\right\}=-\frac{\lambda}{2}\vec{n}$$

$\therefore$ 即是椭圆法线

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33. (道格拉斯生产模型) 设生产利润 p 依赖于投入要素 x，y 和 z (单位价格分别为 a，b 和 c)，满足 $p=kx^{\alpha}y^{\beta}z^{\gamma}$，$\alpha,\beta,\gamma>0$，且 $\alpha+\beta+\gamma=1$，其中 x，y，z 服从价格约束 $ax+by+cz=d$，试确定 x，y 和 z，使 p 取得最大值.

解：$F=ax+by+cz-d$

$L=p+\lambda F$，构造拉格朗日函数

$$\begin{cases} L_x'=k\cdot y^{\beta}\cdot z^{\gamma}\cdot\alpha\cdot x^{\alpha-1}+\lambda a=0 \\ L_y'=k\cdot x^{\alpha}\cdot z^{\gamma}\cdot\beta\cdot y^{\beta-1}+\lambda b=0 \\ L_z'=k\cdot y^{\beta}\cdot x^{\alpha}\cdot\gamma\cdot z^{\gamma-1}+\lambda c=0 \\ F=0 \end{cases}$$

可得 $\begin{cases} \frac{y}{x}\cdot\frac{\alpha}{\beta}=\frac{a}{b} \\ \frac{z}{x}\cdot\frac{\alpha}{\gamma}=\frac{a}{c} \end{cases} \Rightarrow \begin{cases} y=\frac{a}{b}\cdot\frac{\beta}{\alpha}\cdot x \\ z=\frac{a}{c}\cdot\frac{\gamma}{\alpha}\cdot x \end{cases}$

代入F有

$$ax+\frac{1}{\alpha}\cdot a(\beta+\gamma)\cdot x=d$$

$$ax+\frac{1}{\alpha}a(1-\alpha)\cdot x=d$$

$$x=\alpha\cdot\frac{d}{a}$$

同理可得 $y=\beta\frac{d}{b}$，$z=\gamma\cdot\frac{d}{c}$

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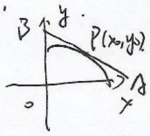
$4y, -8x$

$36x \quad 16y \quad 9x^2+4y^2=72.$

34. 在曲线$18x^2+8y^2=144\,(x\geq 0,\ y\geq 0)$上求一点，使曲线在此点处的切线与曲线及$x$轴，$y$轴所围成的图形具有最小面积.

解：



如图，$F=18x^2+8y^2-144$

对x求偏导.

$F_x'=36x$

$F_y'=16y.$

$\therefore P(x_0,y_0)$处切线方程.

$36x_0(x-x_0)+16y_0(y-y_0)=0$

$\therefore 9x_0\cdot x+4y_0y=72.$

$\therefore A(m,0)\quad B(0,n)$

$\angle m=\dfrac{8}{x_0},\ n=\dfrac{18}{y_0}$

$\therefore S_{\triangle AOB}=\dfrac{144}{x_0y_0}$

即x_0y_0最大时，$S_{\triangle AOB}$最小

构造拉格朗日函数

$L=xy+\lambda(18x^2+8y^2-144)=0$

$$\therefore\begin{cases}L_x'=y+\lambda\cdot 36x=0\\ L_y'=x+\lambda\cdot 16y=0\\ \dfrac{x^2}{8}+\dfrac{y^2}{18}=1\end{cases}$$

又$\because x>0,\ y>0$

$$\therefore\begin{cases}x=2\\ y=3\\ \lambda=\dfrac{3}{2}\end{cases}$$

由实际意义知，$(2,3)$为极大值点

故点$(2,3)$满足题意.

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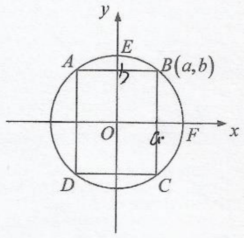
35. 如图，在半径为 1 的圆内有一边长分别为$2a$和$2b$的内接矩形$ABCD$，设弓形$AEBA$绕x轴旋转所得的立体体积为u，弓形$BFCB$绕y轴旋转所得的立体体积为v，问当a，b为何值时，u^2+v^2最小？



解：$x^2+y^2=1$

$$u=\int_{-a}^{a}\pi y^2dx-\pi b^2\cdot a.$$

$$=\pi\cdot(2a-\frac{2}{3}a^3)-2\pi b^2a$$

$$\cancel{=2\pi[(1-b^2)\cdot a-\frac{2}{3}a^3]}$$

$$=2\pi(1-b^2)a-\frac{2}{3}a^3$$

$$=\frac{4}{3}\pi a^3$$

同理可得 $v=\frac{4}{3}\pi b^3$.

∴可构造拉格朗日函数

$$L=a^6+b^6+\lambda(a^2+b^2-1)=0$$

$$\begin{cases}L_x'=6a^5+2\lambda a=0\\ L_y'=6b^5+2\lambda b=0\\ x^2+y^2=1.\end{cases}$$

∴ $a=b=\sqrt[4]{-\frac{\lambda}{3}}$

∴ $\begin{cases}a^2+b^2=1\\ a=b\end{cases}$

∵$a>0$, $b>0$

∴ $a=b=\frac{\sqrt{2}}{2}$.

由实际意义知$(\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2})$为极小值点

∴ $a=b=\frac{\sqrt{2}}{2}$.

或：

同理 由均值定理

$u^2+v^2\geq 2uv$

当$u=v$时，等号成立.

∴ $a=b$ 及 $a^2+b^2=1$

可得 $a=b=\frac{\sqrt{2}}{2}$.

36. 已知 x，y，z 为实数，且 $e^x+y^2+|z|=3$，求证：$e^x y^2|z|\le 1$.

解：令 $u=e^x$，$v=y^2$，$w=|z|$

$\therefore u+w+v=3$

目标函数 $F=uvw$.

$\therefore$ 构造拉格朗日函数

$L=F+\lambda(u+w+v-3)$

$$\begin{cases} L_u'=vw+\lambda=0 \\ L_v'=uw+\lambda=0 \\ L_w'=uv+\lambda=0 \\ u+v+w=3 \end{cases}$$

$\therefore$ 可得 $u=v=w$

$\therefore u=v=w=1$

由几何意义知 $F(1,1,1)$ 为极大值点

$\therefore F_{max}=1$

$\therefore F\le F_{max}=1$

故 $uvw\le 1$

$e^x\cdot y^2|z|\le 1$.

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